



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 1/8/2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : ME F316
Course Title : MANUFACTURING MANAGEMENT
Instructor-in-Charge : GIRISH KANT GARG
Tutorial Instructors : Abhishek Tandey, Girish Kant Garg, Shubham Shivanand Khot, Preeti

1. Course Description:

Introduction to manufacturing systems, forecasting, life cycle concepts, facility location and layout planning, aggregate and batch production planning, scheduling, inventory control, material requirement planning, and enterprise resource planning, just-in time and lean manufacturing, total quality management, supply chain management and Industry 4.0.

2. Scope and Objective of the Course:

The objective of this course is to impart important decision making processes and analytical tools in design, planning and control of manufacturing / service processes. At the end of the course the students shall be able to establish routes and schedules for work that will ensure the optimum utilization of men, materials and machines in a manufacturing / services.

- Familiarize fundamental concepts in production / operations management
- Understand the decision making process in design, planning and control of manufacturing / service systems
- Develop skills for decision making in conversion process / manufacturing systems

3. Text Books:

Russell R. S. & Taylor B. W., "Operations Management", International Student Version, 7/e, John Wiley and Sons (Asia) Pte. Ltd., 2011.

4. Reference Books:

- R1. Chase R.B., Aquilano N.J., and Jacobs F.R., "Operation Management for Competitive Advantage", 9th Edition, Tata McGraw-Hill, Delhi, 2002.
- R2. Krajewski L. J., and Ritzman L.P., "Operations Management: Strategy and Analysis", 6th Edition, Pearson Education Asia, India, 2003.
- R3. Wild R., Operations Management, 6th Ed., Thomson Learning, 2003.

5. Course Plan:

Module No.	Lecture Session	Reference	Learning outcomes
Introduction to Operations Management	1-2	T1	Role and importance of operations management in a plant. The evolution of manufacturing management.
Layout planning	3-7	T7	Facility layout, basic layouts, design of layouts



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Material Handling Systems	8-9	Class Notes	Analysis of material handling systems
Scheduling	10-15	T17, T9	Loading, sequencing, monitoring, project scheduling using CPM/PERT
Forecasting	16-17	T12	Role of forecasting, components of forecasting demand, forecasting methods, forecasting accuracy
Inventory management	18-20	T13	Role of inventory, elements of inventory, EOQ, EBQ models, Inventory classification methods, inventory control systems
Product and process planning	21-23	T4, T6	The product design process, concurrent design, technology in design, QFD, Process planning, analysis and innovation, technology decisions
Aggregate planning	24-26	T14	Operations planning process, quantitative techniques for aggregate planning
Resource planning	27-28	T15	Material requirements planning, master production schedule

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	30	8/10 FN1	Closed Book
Comprehensive Examination	180 Min.	40	1/12 FN	Closed Book
*Tutorials and Quiz	-	20 + 10	Tutorial Class	Open Book

* Best four tutorial test will be considered

7. Chamber Consultation Hour: Wednesday 10th hour (5.00 pm). Chamber No.: 2223-G. Tutorial instructors will announce their consultation hours separately.

8. Notices: All notices will be communicated through Nalanda/Email.

9. Make-up policy: **No Make-up is permissible in Tutorials and Quiz.** Make-up (Mid sem and Compre exam) will be granted only to genuine cases with prior permission.

10. Minimum passing Marks criteria: Scores below 10% of the highest course score or misses any key evaluation components, such as tutorials, midterm tests, or the comprehensive exam, will be awarded an NC grade.

Instructor-in-charge
Course No. ME F316